8.9.3 Effects of improper wheel alignment

1. Vehicle will roll over due to centrifugal force acting at the time of turning
2. Wheels are subjected to vibration
3. Increases the tyre wear
4. Reduce the tyre life
5. Hard steering
6. Vehicle pull on one side

8.10 Difference between Mechanical Steering and Power Steering

Sl No.	Mechanical Steering	Power Steering
1	Mechanical steering is a system in which Mechanical force is used for steering.	Power steering is a system that helps in steering the wheels using some source of power viz Hydraulic, Electric, Air.
2	Resistance to wheel movement is more.	Resistance to wheel movement is less
3	Widely preferred for Low weight vehicles.	Ease of steering in Comparatively heavy weight vehicles.
4	Steering wheel and column, a Mechanical gearbox and pitman arm or a rack and pinion assembly, linkages; steering knuckles and ball joints, and 4. the wheel spindle assemblies.	A hydraulic pump, fluid reservoir, hoses, lines; and either a power assist unit mounted on, or integral with, a power steering gear assembly.
5	Does not Absorbs road shocks.	Absorbs road shocks.
6	Efforts required for steering is high.	Efforts required for steering is less.
7	Very difficult to control directional at the time of emergency and directional stability is difficult to achieve.	Greater safety and controllability under emergency situation.

8.11 Front Axle

Front axle carries the weight of the front part of the vehicle. Front axle includes axle beam stub axles with brake assembly, track rod and stub axle arm to steer and brake the vehicle. To accommodate front engine and to provide stability and safety, the centre portion of the axle is dropped. It is called as front axle.

Construction

In light vehicles, front axles are tubular in section and in heavy commercial vehicles, the front axles are made up of I cross section. In both the ends, stub axles are connected with front axle by means of kingpin and steering track arm.

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